

Objective: Explore different types of biological databases and classify them.

Task: Choose any three databases (Primary, Secondary and Specialized databases)

Identify and describe their data type, format and purpose.

Bioinformatics relies heavily on databases to store, analyze and interpret biological data. The different types of bioinformatics databases are:

PRIMARY DATABASES – They store raw, unprocessed data directly from experimental methods. For example:

PRIMARY DATABASES	DATATYPE	FORMAT	PURPOSE
GenBank	Nucleotide sequences, Annotation, bibliographic and biological information.	Flat file format	Provide and encourage access within scientific community to the most up to date and comprehensive DNA sequence information.
EMBL(European Molecular Biology Laboratory)	Nucleotide sequences, protein sequences and related information.	Flat file format	To perform fundamental research, offer vital service to scientists, train scientist and students and engage in technology transfer and industry relations.
DDBJ(DNA Databank of Japan)	Nucleotide sequence data including WGS (Whole Genome Shotgun), RNA, cDNA, synthetic sequences, environmental sequence data and data from Next Generating Sequencing platforms.	Flat file format	Serves as repository for nucleotide and Amino Acid sequence data and also a key player in INSDC (International Nucleotide Sequence Database Collection).
PDB(Protein Databank)	Stores and shares 3D structure of biological macromolecule including (Protein and Nucleic Acid) along with related information.	Text file format	Serves as central resource for structural information, enabling researchers to study protein structure, understanding their function and develop new technologies.

SECONDARY DATABASES- They are derived from primary databases which focuses on processed or interpreted data such as annotations, motifs or protein families. For example:

SECONDARY DATABASES	DATATYPE	FORMAT	PURPOSE
UniProt	Freely accessible database of protein sequences and functional information.	Flat text, FASTA, XML, RDF and GFF.	It is used for storing and interconnecting information from large and disparate sources.
Pfam	It is a curated database of protein families, domains, repeats, motifs, coiled-coil and disorders.	It is an ASCII file.	It aids in genome annotation, protein structure and functional analysis, and evolutionary studies.
PROSITE	It is a database of documentation entries that describe protein domains, families, and functional sites derived from multiple sequence alignment.	Three ASCII text files, including a DAT file for motif matches.	It serves to identify protein families, domains, and functional sites by using patterns and profiles.
PRINTS (Protein Family Fingerprints)	It is a collection of protein fingerprints, which may be used to assign family and functional attributes to uncharacterized sequence.	PDF, JPEG, TIFF and EPS.	It serves as diagnostic resource for identifying and characterizing newly determined protein sequences.
SWISS-PROT	It is a curated protein sequence database with a high level of annotations such as functions, domain structure, post-translational modifications and variants.	Flat file format.	To provide a curated, high quality and non-redundant protein sequence database with detailed annotations.

SPECIALISED DATABASES- These databases are designed to serve a specific research interest, concentrating on a particular organism, datatype and biological process. For example:

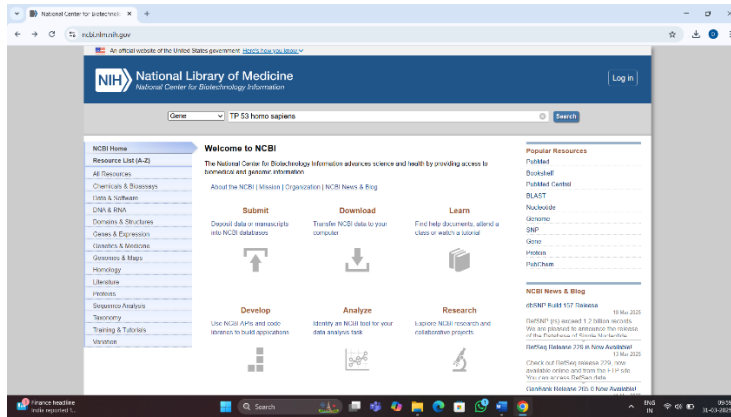
SPECIALIZED DATABASES	DATATYPE	FORMAT	PURPOSE
Flybase (Drosophila)	It is a web portal for genetic and genomic information on <i>Drosophila melanogaster</i> and related species, with data types including genes, alleles,	XML, ASCII flat files, FASTA, GFF and GTF files.	It has information on gene sequences, alleles, mutations, genome and amino acid sequences of <i>Drosophila melanogaster</i> .

	phenotypes, aberrations, sequences, stocks, images, and more		
HIV sequence database	It encompasses both nucleic acid and amino acid sequences, crucial for understanding the virus's structure, evolution, and drug resistance and is used to classify HIV into groups and subtypes.	FASTA format.	It helps to collect, curate and analyse HIV sequences, aiding in research on drug resistance, vaccine development and understanding viral evolution.
RDP(Ribosomal Database Project)	It primarily deals with aligned and annotated ribosomal RNA sequences, derived phylogenetic trees, and related analysis tools.	GenBank-formatted files.	It provides the research community with aligned and annotated rRNA gene sequence data, analysis services, and tools, including a phylogenetically consistent taxonomic framework.
KEGG (Kyoto Encyclopedia of Genes and Genomes)	It is an ontology database containing hierarchical classifications of various entities including genes, proteins, organisms, diseases, drugs, and chemical compounds.	KGML (KEGG Markup Language).	It is used for understanding high-level functions and utilities of the biological system, such as the cell, the organism and the ecosystem, from molecular-level information, especially large-scale molecular datasets generated by genome sequencing and other high-throughput experimental technologies.
REACTOME	It is an open-source, open access, manually curated and peer-reviewed pathway database.	BioPAX, PSI-MITAB, SBML, and tab-delimited text files.	It provides tools for the visualization, interpretation and analysis of pathway knowledge to support basic and clinical research, genome analysis, modelling, systems biology and education.

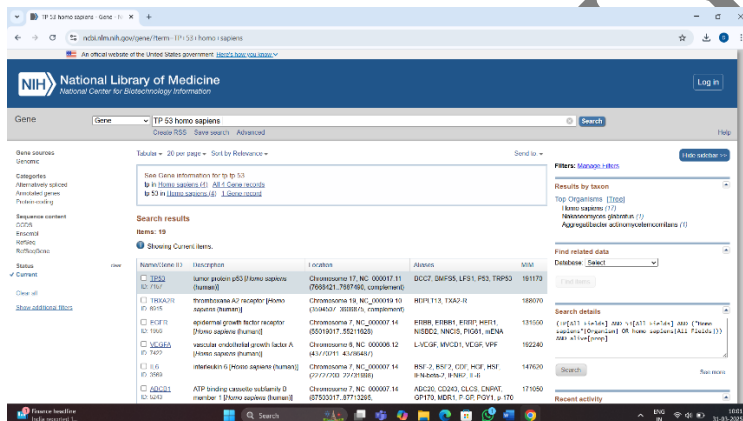
Retrieve one dataset from each and explain how to extract relevant information.

1.PRIMARY DATABASE(GenBank)

- Go to the NCBI website.
- Select gene from drop down menu and type your gene of interest on search bar (TP (Tumor Protein)53 *Homo sapiens*).



- Select the file you want to open, here we are focusing on studying the gene sequence of *Homo sapiens* .



- You can then analyze the summary, genomic content, expression, bibliography etc.

TP53 tumor protein p53 [Homo sapiens (human)]

Gene ID: 7157, updated on 26-Mar-2025

Summary

Official Symbol: TP53, reviewed by HGNC

Official Full Name: tumor protein p53, provided by HGNC

Primary source: HGNC:HGNC:11998

See related: Ensembl: ENSG00000141510, MIM: 191173, AllianceGenome: HGNC:11998

Gene type: protein coding

RefSeq status: REVIEWED

Organism: Homo sapiens

Lineage: Eukaryota, Metazoa, Chordata, Craniata, Vertebrata, Euteleostomi, Mammalia, Eutheria, Euarchontoglires, Primates, Haplorhina, Catarrhini, Hominoidea, Homo

Also known as: P53, BCG7, LFS1, BMF-S5, TRP53

Summary: This gene encodes a tumor suppressor protein containing transcriptional activation, DNA-binding, and oligomerization domains. The encoded protein responds to diverse cellular stresses to regulate expression of target genes, thereby inducing cell cycle arrest, apoptosis, senescence, DNA repair, or changes in metabolism. Mutations in this gene are associated with a variety of human cancers, including hereditary cancers such as Li-Fraumeni syndrome. Alternative splicing of this gene and the use of alternate promoters result in multiple transcript variants and isoforms. Additional isoforms have also been shown to result from the use of alternate translation initiation codons from identical transcript variants (PMIDs: 12032546, 20937277). [provided by RefSeq, Dec 2016]

Ubiquitous expression in spleen (RPKM 13.2), lymph node (RPKM 13.1) and 25 other tissues [View data](#)

Expression: [tissue](#) [all](#)

Orthologs: [Try the new Gene table](#)
[Try the new Transcript table](#)

Genomic context

Location: 17p13.1 [See TP53 in Genome Data Viewer](#)

Exon count: 13

Annotation release	Status	Assembly	Chr	Location
RS_2024_08	current	GRCh38.p14 (GCF_000001405.40)	17	NC_000017.11 (7666421..7687400, complement)
RS_2024_08	current	T2T-CHM13v2.0 (GCF_009914755.1)	17	NC_060941.1 (7572544..7591594, complement)
RS_2024_09	previous assembly	GRCh37.p13 (GCF_000001405.25)	17	NC_000017.10 (7571739..7590808, complement)

Genomic regions, transcripts, and products

Genomic Sequence: NC_000017.11 Chromosome 17 Reference GRCh38.p14 Primary Assembly

Go to nucleotide: [Graphics](#) [FASTA](#) [Getlink](#)

Tools: [Track](#) [Download](#) [Configure tracks](#)

Full track in PMc

Full track in PMc, nucleotide

Functional Class

GAP

Gene neighbors

Genes with a similar H3K4me3 profile

Genome Browsers

[Genome Data Viewer](#)

[Variation Viewer \(GRCh37.p13\)](#)

[Variation Viewer \(GRCh38\)](#)

[Ensembl](#)

[UCSC](#)

Related information

[3D structures](#)

[BioAssay by Target \(List\)](#)

[BioAssay by Target \(Summary\)](#)

[BioAssay by Gene target](#)

[BioAssays: RNAi Target Active](#)

[BioAssays: RNAi Target Tested](#)

[BioProjects](#)

[BioSystems](#)

[Books](#)

[CCDS](#)

[ClinVar](#)

[Conserved Domains](#)

[dbVar](#)

[Full text in PMC](#)

[Full text in PMC, nucleotide](#)

[Functional Class](#)

[GAP](#)

[Gene neighbors](#)

[Genes with a similar H3K4me3 profile](#)

The image displays three screenshots of the NCBI Gene database interface for the TP53 gene (Gene ID: 7157).

Top Screenshot: Genomic Context

- Genomic Sequence:** NC_000017.11 Chromosome 17 Reference GRCh38.p14 Primary Assembly
- Gene Structure:** Shows exons (green boxes) and introns (lines) for TP53. The gene is located on chromosome 17 at approximately 7,680,000 to 7,680,000 coordinates.
- Annotations:** Includes RefSeq, Ensembl, and UniProt annotations. The RefSeq entry is NM_001258696.2.

Middle Screenshot: Expression Data

- Expression:** Tissue-specific circular RNA induction during human fetal development.
- Bar Chart:** Shows RPKM values for various tissues. The highest expression is observed in testis and ovary.

Bottom Screenshot: Bibliography and GeneReviews

- Bibliography:** Lists related articles in PubMed, including studies on TP53 protein expression, mutations, and its role in cancer.
- GeneReviews:** Provides information on the clinical significance of TP53 mutations, including their association with various cancers and the availability of genetic testing.

- You can then explore the genomic sequence from GenBank option of Genomic Sequence to find the whole gene sequence, CDS regions, messenger RNA sequences, etc. , which helps in understanding gene function, variations, comparative study, gene discovery, develop personalized medicine and discover new treatments.

Homo sapiens chromosome 17, GRCh38.p14 Primary Assembly

NCBI Reference Sequence: NC_000017.11

[FASTA](#) [Graphics](#)

LOCUS NC_000017 19070 bp DNA linear CON 26-AUG-2024

DEFINITION Homo sapiens chromosome 17, GRCh38.p14 Primary Assembly.

ACCESSION NC_000017 REGION: complement(7668421..7687490)

VERSION NC_000017.11

DBLINK BioProject: [PRJNA168](#)
Assembly: [GCF_000001495.40](#)

KEYWORDS RefSeq

SOURCE Homo sapiens (human)

ORGANISM [Homo sapiens](#)
Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini; Catarrhini; Hominidae; Homo.

REFERENCE 1 (bases 1 to 19070)

AUTHORS Zody, M.C., Garber, M., Adams, D.J., Sharpe, T., Harrow, J., Lupski, J.R., Nicholson, C., Searle, S.M., Wilming, L., Young, S.K., Aboueleil, A., Allen, N.R., Bi, W., Bloom, T., Borowsky, M.L., Bugalter, B.E., Butler, J., Chang, J.L., Chen, C.K., Cook, A., Corum, B., Cuomo, C.A., de Jong, P.J., DeCaprio, D., Dewar, K., Fitzgerald, M., Gilbert, J., Gibson, R., Gnerre, S., Goldstein, S., Grafham, D.V., Grocock, R., Hafez, N., Hagopian, D.S., Hart, E., Norman, C.H., Humphray, S., Jaffe, D.B., Jones, M., Kamal, M., Khodiyar, V.K., LaButti, K., Laird, G., LeHocqy, J., Liu, X., Lokysang, T., Loveland, J., Lui, A., Macdonald, P., Major, J.E., Matthews, L., Mauceli, E., McCarroll, S.A., Nihalev, A.H., Hodge, J., Nguyen, C., Nicol, R., O'Leary, S.B., Osogawa, K., Schwartz, D.C., Shaw-Smith, C., Stankiewicz, P., Steward, C., Swarbreck, D., Venkataraman, V., Whittaker, C.A., Yang, X., Zimmer, A.R., Bradley, A., Hubbard, T., Birren, B.W., Rogers, J., Lander, E.S. and Nusbaum, C.

TITLE DNA sequence of human chromosome 17 and analysis of rearrangement in the human lineage

JOURNAL Nature 440 (7887), 1045-1049 (2006)

PUBMED 16625196

REFERENCE 2 (bases 1 to 19070)

CONSRTH International Human Genome Sequencing Consortium

TITLE Finishing the euchromatic sequence of the human genome

JOURNAL Nature 401 (7011), 931-945 (2000)

Change region shown
☐ Whole sequence (abbreviated view)
☒ Selected region
from: 7668421 to: 7687490
[Update View](#)

Customize view
☐ Abbreviated view
☒ Customize
Basic Features
☐ All features
☒ Gene, RNA, and CDS features only
Display options
☒ Show reverse complement
☐ Show gap features
[Update View](#)

Analyze this sequence
[Run BLAST](#)
[Pick Primers](#)
[Highlight Sequence Features](#)
[Find in this Sequence](#)

Related information
[Assembly](#)
[BioProject](#)
[Protein](#)
[PubMed](#)
[Taxonomy](#)

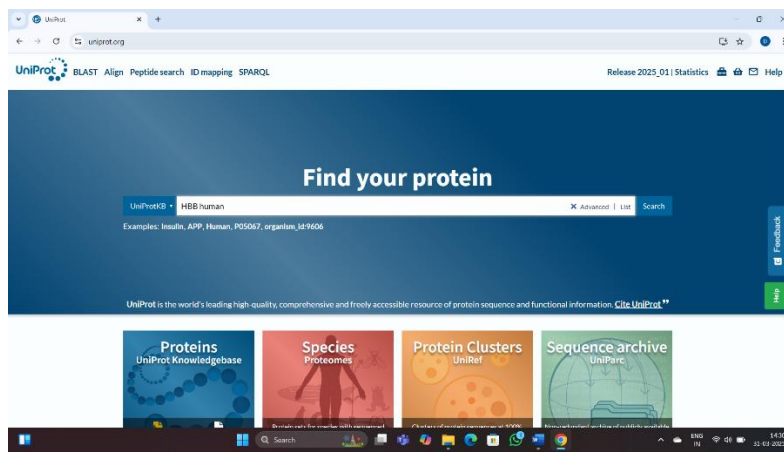
- You can then get the FASTA sequence from the GenBank for sequence analysis with improved accuracy in genomics, transcriptomics, proteomics studies.

FASTA

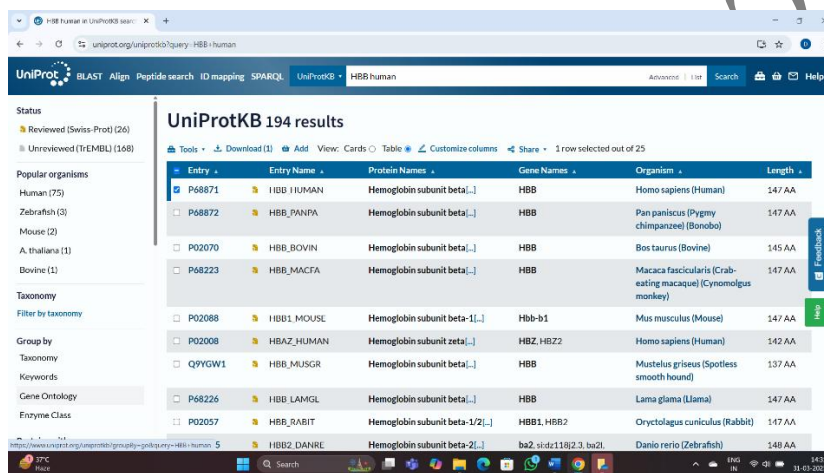
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1921 gaagtagag caggaatc gctgagccc agagagcga agttgagtg agcagatc
1981 gtgccaac actccagctt ggcgacag agcgaactc cgtctcaaa aagaggggg
2041 aatcatatc tatgtgcta ttttttgcg gctttgttc ttaatgtac tgcctgac
2101 tgcctcatg tgtatatac agatttttc tcttttcat ttagtagat ccacgattg
2161 tatatcgtc cttttgagc cttttttag ttgttccat ttgcgttat gaataaagc
2221 tgcataaac attctgtac attctttttt gtagcatat gtttcgtgt ttcttggaga

2. SECONDARY DATABASE (UniProt)

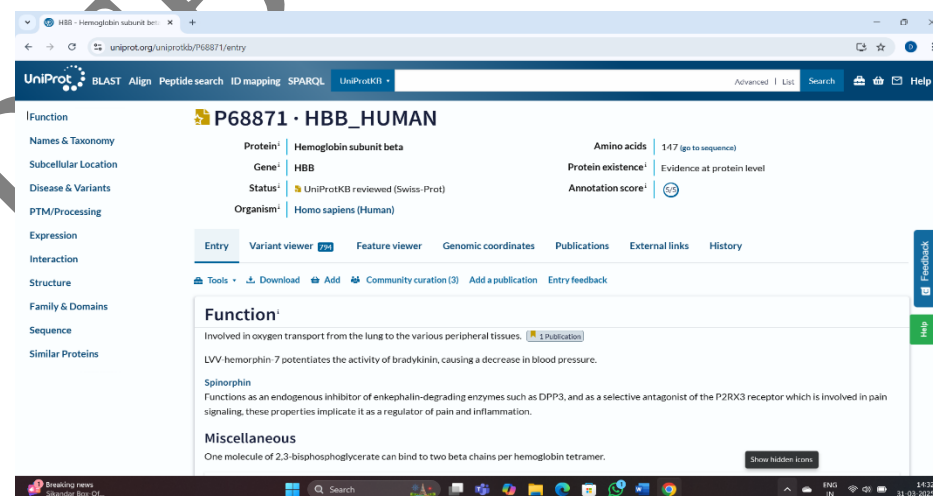
- Go to the UniProt website and search your protein of interest (HBB (Haemoglobin subunit Beta) in humans).



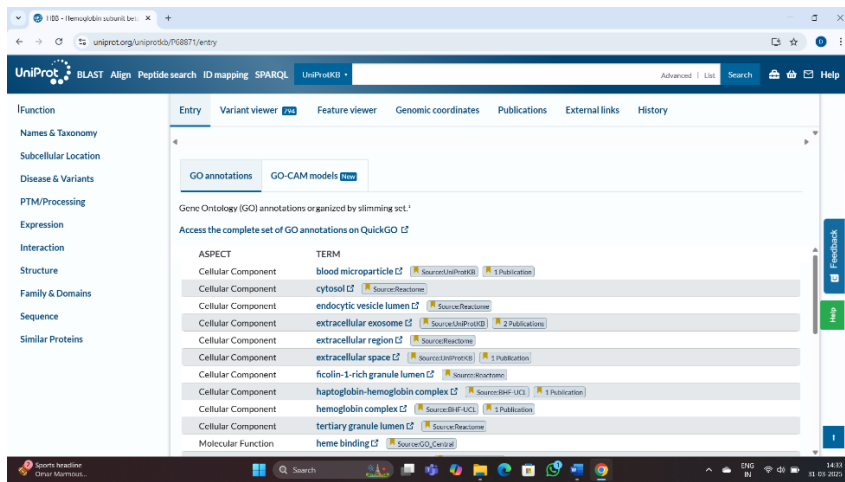
- Select the protein sequence you want to study.



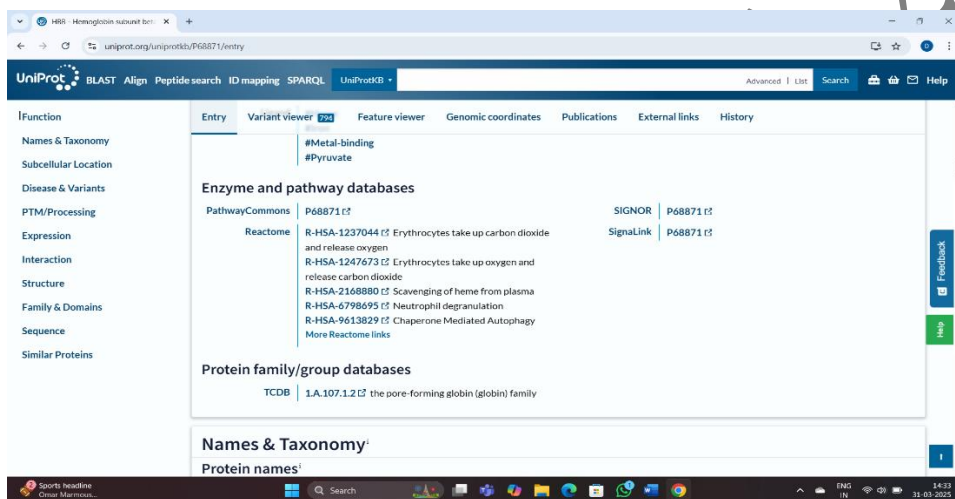
- Under function, we can study how protein plays an important role in transporting oxygen, how mutations lead to genetic disorder and how it is highly conserved in different species.



- Gene ontology provides information on the transportation of oxygen, biosynthesis of haemoglobin, differentiation of RBCs, how HBB is associated with RBC membrane and binds oxygen.



- Enzyme and pathway database tells about the globin protein and its oxygen binding activity, the oxygen transport pathway, the RBC differentiation pathway, interactions with other globin proteins and molecules.



- Names and taxonomy gives insights of its alternative names, its classification, accession number, organism it is obtained from and the lineage.

UniProtKB | BLAST | Align | Peptide search | ID mapping | SPARQL | UniProtKB | Search | Help

Function | Entry | Variant viewer | Feature viewer | Genomic coordinates | Publications | External links | History

Names & Taxonomy

Protein names

Recommended name	Hemoglobin subunit beta
Alternative names	Beta-globin Hemoglobin beta chain
Cleaved into 2 chains	LVV-hemorphin-7 Spinorphin

Gene names

Name	HBB
------	-----

Community curation (3)

Organism names

Taxonomic identifier	9606 (NCBI)
Organism	Homo sapiens (Human)
Taxonomic lineage	cellular organisms > Eukaryota (eucaryotes) > Opisthokonta > Metazoa (metazoans) > Eumetazoa > Bilateria > Deuterostomia > Chordata (chordates) > Craniata > Vertebrata (vertebrates) > Gnathostomata (jawed vertebrates) > Teleostomi > Euteleostomi (bony vertebrates) > Sarcopterygii > Dipsosauromorpha > Tetrapoda (tetrapods) > Amniota (amniotes) > Mammalia (mammals) > Theria > Eutheria (placental) > Boreoeutheria > Euarchontoglires > Primates > Haplorhini > Simiiformes > Catarrhini > Hominoidea (apes) > Hominidae (great apes) > Homininae > Homo

Accessions

Primary accession	P68871
Secondary accessions	A4GX73 B2ZUE0 P02023 Q13852 Q14481 More accessions

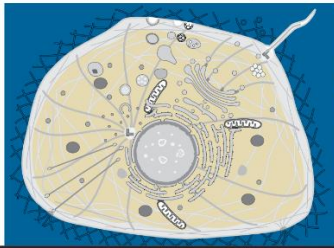
- Subcellular location specifies the places the gene is located so that we can understand the prime functions and repercussion of it's possible mutations. The gene ontology gets further explained here, as we find the exact location to study the functions.

UniProtKB | BLAST | Align | Peptide search | ID mapping | SPARQL | UniProtKB | Search | Help

Function | Entry | Variant viewer | Feature viewer | Genomic coordinates | Publications | External links | History

Subcellular Location

UniProt Annotation: Secreted
Protein located outside the cell membrane(s)



Subcellular Location

- blood microparticle
- cytosol
- endocytic vesicle lumen
- extracellular exosome
- extracellular region
- extracellular space
- ficollin-1-rich granule lumen
- haptoglobin-hemoglobin complex
- hemoglobin complex
- tertiary granule lumen

[Complete GO annotation on QuickGO](#)

- Disease and variants, specifies about the different genetic disorders like Beta Thalassaemia, Sickle cell anaemia, HBBAN; also about the variants of the gene which

could to different forms of disorders, different mutations leading to a genetic disorder and the mechanism of disease which helps in genetic counselling.

The image displays three screenshots of the UniProt database interface, specifically focusing on the entry for HBB (Hemoglobin subunit beta).

Top Screenshot: Disease & Variants

- Function:** Entry, Variant viewer, Feature viewer, Genomic coordinates, Publications, External links, History.
- Names & Taxonomy:** Subcellular Location, Disease & Variants, PTM/Processing, Expression, Interaction, Structure, Family & Domains, Sequence, Similar Proteins.
- Disease & Variants:**
 - Involvement in disease¹**
 - Heinz body anemias (HEIBAN)**
 - Note:** The disease may be caused by variants affecting the gene represented in this entry.
 - Description:** Form of non-spherocytic hemolytic anemia of Dacie type 1. After splenectomy, which has little benefit, basophilic inclusions called Heinz bodies are demonstrable in the erythrocytes. Before splenectomy, diffuse or punctate basophilia may be evident. Most of these cases are probably instances of hemoglobinopathy. The hemoglobin demonstrates heat lability. Heinz bodies are observed also with the Ivemark syndrome (asplenia with cardiovascular anomalies) and with glutathione peroxidase deficiency.
 - See also:** MIM:140700 (c)
 - Beta-thalassemia (B-THAL)**
 - Note:** The disease is caused by variants affecting the gene represented in this entry.
 - Description:** A form of thalassemia. Thalassemias are common monogenic diseases occurring mostly in Mediterranean and Southeast Asian populations. The hallmark of beta thalassemia is an imbalance in globin chain production in the adult HbA molecule. Absence of beta chain causes beta(0) thalassemia, while reduced amounts of detectable beta globin causes beta(+)-thalassemia. In the severe forms of beta-thalassemia, the excess alpha globin chains accumulate in the developing erythroid precursors in the marrow. Their deposition leads to a vast increase in erythroid apoptosis that in turn causes ineffective erythropoiesis and severe microcytic hypochromic anemia. Clinically, beta-thalassemia is divided into

Middle Screenshot: Features

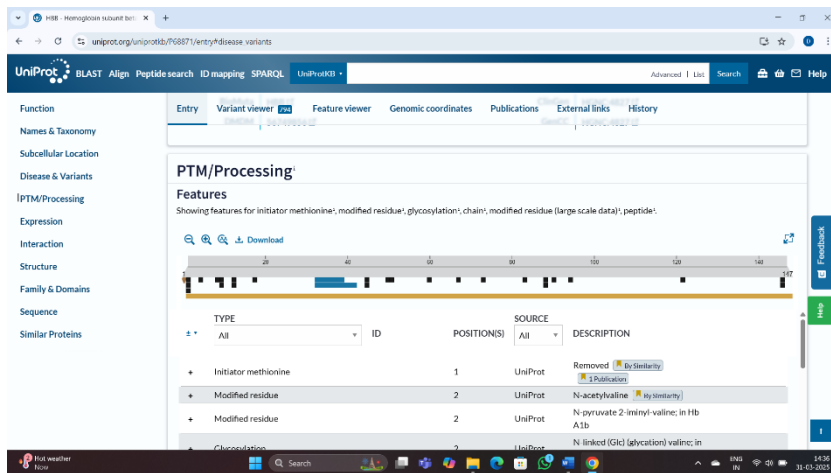
- Showing features for natural variant¹:**
- Download:** [Download]
- Table:**

TYPE	ID	POSITION(S)	DESCRIPTION
Natural variant	VAR_002856	2	in Raleigh; O ₂ affinity down; dbSNP:rs33949930 (c)
Natural variant	VAR_002857	3	in Graz; dbSNP:rs33983205 (c) [1 Publication]
Natural variant	VAR_002858	3	in Okayama; O ₂ affinity up; dbSNP:rs713040 (c)
Natural variant	VAR_002859	3	in Deer Lodge; O ₂ affinity up; dbSNP:rs33983205 (c)
Natural variant	VAR_002860	3	in Fukuoka; dbSNP:rs35906307 (c)
Natural variant	VAR_002861	6	in Warwickshire; dbSNP:rs34769005 (c)

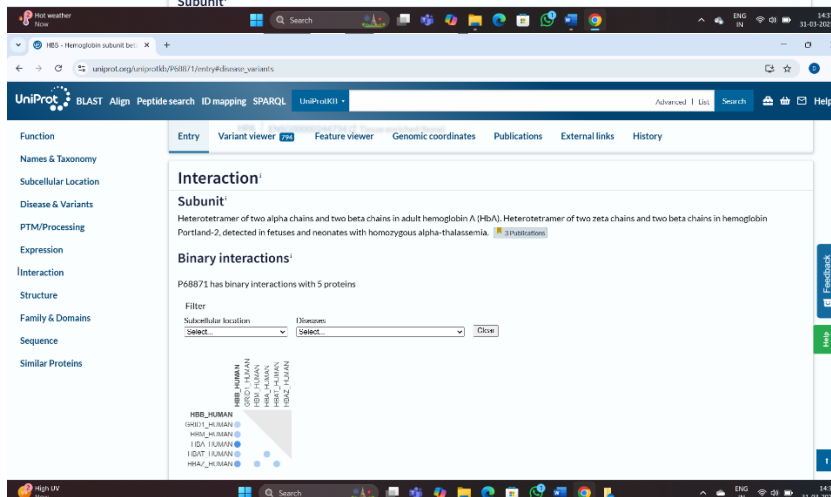
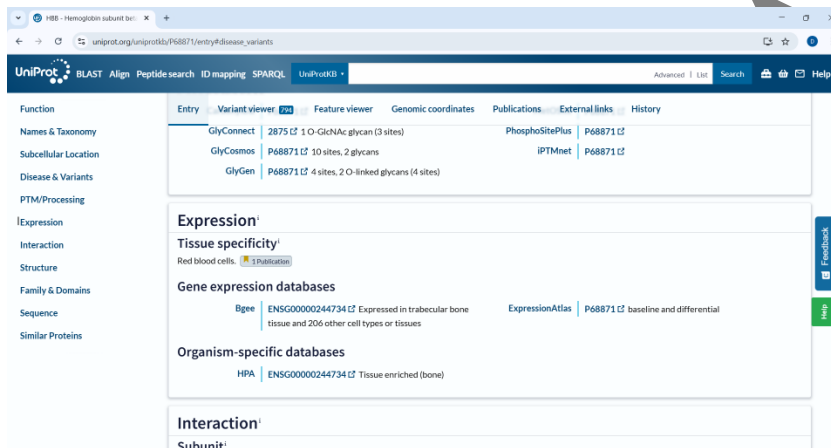
Bottom Screenshot: Variants

- Download:** [Download]
- Filter Consequence:**
 - ☒ Likely pathogenic or pathogenic
 - ☒ Predicted consequence
 - ☒ Likely benign or benign
 - ☒ Uncertain significance
- Filter Provenance:**
 - ☒ UniProt reviewed
 - ☒ ClinVar
 - ☒ Large scale studies
- Sequence:** A detailed view of the HBB protein sequence (147 amino acids) with various variants highlighted by colored dots (red, green, blue, yellow) indicating different types of mutations.

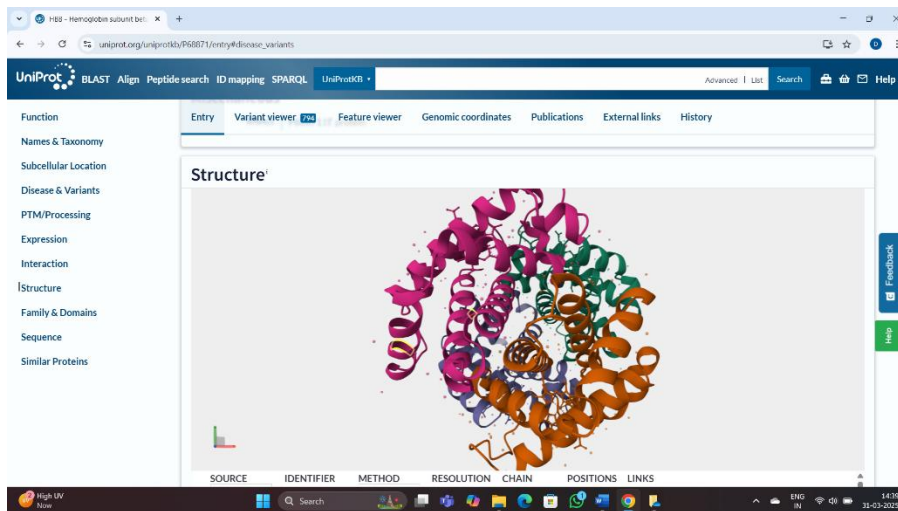
- Post translational modification helps us understand the regulation of the function in normal state and the effects if the modifications are absent, the interactions could be hampered, and the protein would lose stability. It helps discover biomarkers for disease diagnosis.



- The tissue specific expression tells about the transcription factors and proteins it is associated with, and the interactions which leads to the expression of the functions.



- Structure classifies the size, exonic and intronic regions, promoter sequences, regulatory elements, helix diameter, transcription start site to understand mutations.



- It is a member of globin gene family, which characterizes it's functional and protein domains.

UniProt BLAST Align Peptide search ID mapping SPARQL UniProtKB Advanced | List Search Help

Function Entry Variant viewer Feature viewer Genomic coordinates Publications External links History

Names & Taxonomy Subcellular Location Disease & Variants PTM/Processing Expression Interaction Structure Family & Domains Sequence Similar Proteins

Family & Domains

Features

Showing features for domain:

Download

1 20 40 60 80 100 120 140 147

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147

TYPE	ID	POSITION(S)	DESCRIPTION
Domain	3-147		Globin PROSITE-ProRule Annotation

Sequence similarities

Belongs to the globin family. PROSITE-ProRule Annotation

Phylogenomic databases

Database	Accession
GeneTree	ENSGT00940000163476
HOGENOM	CLU_003827_10_0_1
OrthoDB	9886081at2759
PhyloDB	P68871

- Sequence helps classify point, frameshift, deletion or other mutations which leads to disorders.

UniProt BLAST Align Peptide search ID mapping SPARQL UniProtKB Advanced | List Search Help

Function Entry Variant viewer Feature viewer Genomic coordinates Publications External links History

Names & Taxonomy Subcellular Location Disease & Variants PTM/Processing Expression Interaction Structure Family & Domains Sequence Similar Proteins

Sequence

Sequence status: Complete

See also: sequence in UniParc or sequence clusters in UniRef

Tools: Download Add Highlight Copy sequence

Length: 147
Mass (Da): 15,998

Last updated: 2007-01-23 v2
MD5 Checksum: 209D686939D08BD1EA0893A8150140E2

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147

Computationally mapped potential isoform sequences

There are 3 potential isoforms mapped to this entry

Entry	Entry name	Gene name	Length
1			

- Polymorphism tells how SNPs, insertion and deletion, copy number variations could lead to disorder. Sequence similarity helps identify the conserved sequences, homology among species and other proteins and phylogenetic studies.

The top screenshot displays the UniProt entry for P68871 (Hemoglobin subunit beta). The 'Polymorphism' section describes genetic variations in HBB, such as Hb S (Hb S), which are associated with resistance to malaria. It also lists keywords like #3D-structure, #Direct protein sequencing, #Proteomics identification, and #Reference proteome. The 'Sequence databases' section shows links to CCDS, PIR, and RefSeq.

The bottom screenshot shows the 'Similar Proteins' section for the same entry. It lists six similar proteins with their UniProt IDs, names, organisms, and lengths. The proteins are: Hemoglobin subunit beta (P68871) from Pan paniscus (147 aa), Hemoglobin subunit beta (P68871) from Pan troglodytes (147 aa), Hemoglobin subunit beta (P68871) from Culex pipiens (147 aa), Globin A1 (P68871) from Pan troglodytes (147 aa), HBB isoform 1 (P68871) from Pan troglodytes (147 aa), and Beta-globin (P68871) from Homo sapiens (147 aa).

3. SPECIALIZED DATABASE (KEGG)

- Go to the KEGG database and enter gene of interest (CFTR ; Cystic Fibrosis Transmembrane Conductance Regulator) whose pathway you want to study.

KEGG PATHWAY Database

Wiring diagrams of molecular interactions, reactions and relations

KEGG2 PATHWAY BRITTE MODULE KO GENES COMPOUND NETWORK DISEASE DRUG

Select prefix: map | Organism: CFTR | Enter keywords: | Go | Help

[New pathway maps | Update history]

Pathway Maps

KEGG PATHWAY is a collection of manually drawn pathway maps representing our knowledge of the molecular interaction, reaction and relation networks for:

1. Metabolism
 - Global/overview
 - Carbohydrate
 - Energy
 - Lipid
 - Nucleotide
 - Amino acid
 - Other amino
 - Glycan
 - Cofactor/vitamin
 - Terpenoid/PP
 - Other secondary metabolite
 - Xenobiotics
 - Chemical structure
2. Genetic Information Processing
3. Environmental Information Processing
4. Cellular Processes
5. Organismal Systems
6. Human Diseases
7. Drug Development

The pathway map viewer linked from this page is a part of KEGG Web Apps and contains features of KEGG mapping.

Pathway Identifiers

Each pathway map is identified by the combination of 2-4 letter prefix code and 5 digit number (see KEGG Identifier). The prefix has the following meaning:

- map manually drawn reference pathway/identifiers
- ko reference pathway highlighting KOs
- ec reference metabolic pathway highlighting EC numbers
- rn reference metabolic pathway highlighting reactions
- <org> organism-specific pathway generated by converting KOs to geneIDs
- vg viruses pathway generated by converting KOs to geneIDs
- vx viruses extended pathway generated by including synteny analysis data

and the numbers starting with the following:

- 011 global map (links linked to KOs)
- 012 overview map (links linked to KOs)

- The results shows different pathways the gene is involved in, click on the one you want to study(cyclin AMP signaling pathway).

Pathway Search Result

Pathway Text Search

Number of entries in a page [20] | Hide thumbnail

Items : 1 - 8 of 8

Entry	Thumbnail Image	Name	Description	Object	Legend
map04972		Pancreatic secretion	...f pancreatic HCO3- secretion. Activation of the CFTR Cl- channel and the CFTR-dependent Cl-/HCO3- ex...	... (ATP1B1), K01538 (FYOD2) K04936 (KCNMA1) K05031 (CFTR) K04568 (SCTR) K05742 (SLC3A1) K13852 (SLC4A2) ...	... NHE1 HCO Pancreatic duct cell SLC26A3 ATP Na/K CFTR NHE1 AP2 HCO CADC: CAMP CaII RYR: RANAPR: cAMP: RA...
map04024		cAMP signaling pathway	cAMP is one of the most common and universal second messengers, and its formation is promoted by ade...	... K04456 (AKT) K04371 (ERK) K05852 (PLN) K05031 (CFTR) K02183 (CALM) K06014 (RAPGEF3), K04351 (RAPGEF...	... P 2Na NHE GPCR PKA PI3K Akt cAMP DNA ERK1/2 PLB CFTR CaM Cardiac muscle contraction Epac PI3K-Akt si...

- The description elaborates about the pathway and the pathway map shows how the gene is directly or indirectly regulating various gene expression and functions.

Pathway Search Result x KEGG PATHWAY: map04024 x +

kegg.jp/entry/map04024

KEGG PATHWAY: map04024

Entry map04024 Pathway

Name cAMP signaling pathway

Description cAMP is one of the most common and universal second messengers, and its formation is promoted by adenylyl cyclase (AC) activation after ligation of G protein-coupled receptors (GPCRs) by ligands including hormones, neurotransmitters, and other signaling molecules. cAMP regulates pivotal physiologic processes including metabolism, secretion, calcium homeostasis, muscle contraction, cell fate, and gene transcription. cAMP acts directly on three main targets: protein kinase A (PKA), the exchange protein activated by cAMP (Epac), and cyclic nucleotide-gated ion channels (CNGs). PKA modulates, via phosphorylation, a number of cellular substrates, including transcription factors, ion channels, transporters, exchangers, intracellular Ca^{2+} -handling proteins, and the contractile machinery. Epac proteins function as guanine nucleotide exchange factors (GEFs) for both $Rap1$ and $Rap2$. Various effector proteins, including adaptor proteins implicated in modulation of the actin cytoskeleton, regulators of G proteins of the Rho family, and phospholipases, relay signaling downstream from Rap.

Class Environmental Information Processing; signal transduction

Pathway map04024 cAMP signaling pathway

Show desktop

Trending videos The Most Viewed...

Search

ENG IN 16:32 31-03-2025

Pathway Search Result x KEGG PATHWAY: cAMP signalin: x +

kegg.jp/pathway/map04024

KEGG cAMP signaling pathway - Reference pathway

[Pathway menu | Organism group | Pathway entry | Show description | Download | Help]

Change pathway type

Option

Scale: 56%

Search

ID search

Color

Network

☐ nt06310 CRH-ACTH-cortisol sign

☐ N00324 CRHR-PKA-ACTH sign

☐ nt06323 KISS1-GiRHR-LH/FSH-E

☐ N00885 LHGR-GNAS-PKA sig

☐ N00922 FSHR-GNAS-PKA sign

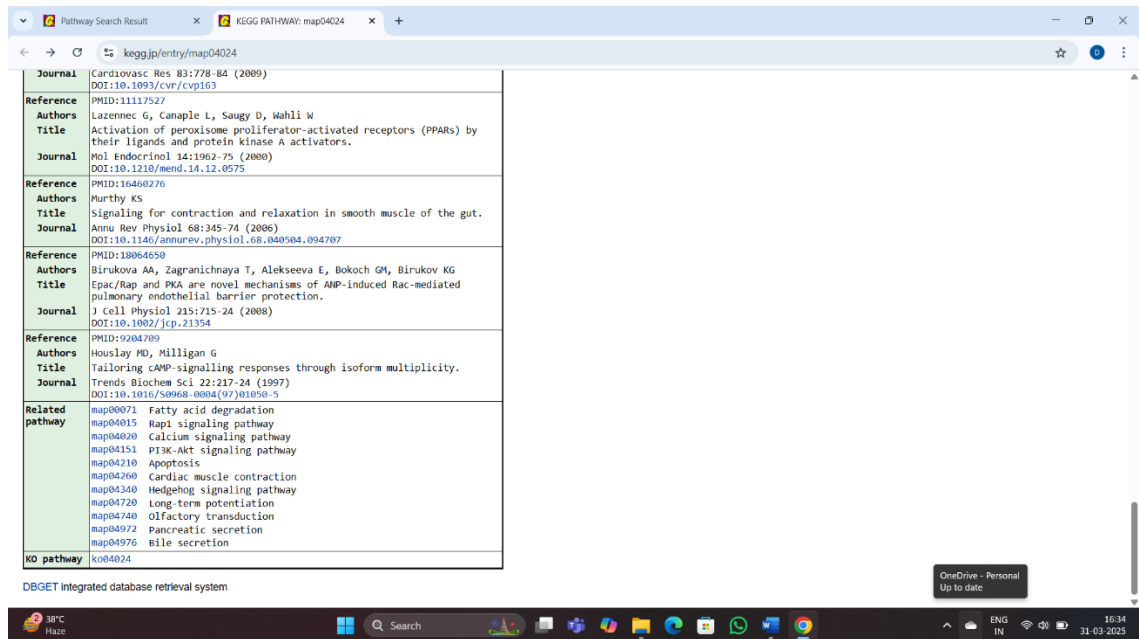
38°C Hazy

Search

ENG IN 16:33 31-03-2025

- The references of various studies or experiments are mentioned which led to creation of the pathway map and in the gene ontology section, similar pathway regulated by

different gene in other organisms can be studied.



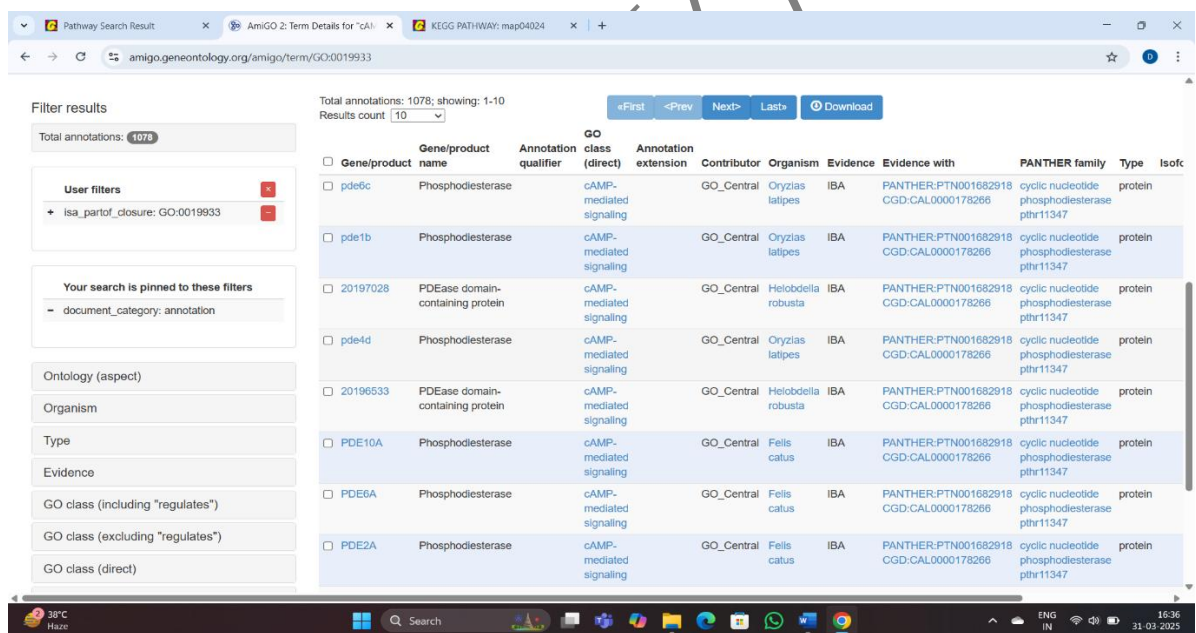
The screenshot shows the KEGG Pathway Search Result for map04024. The page displays a list of references and related pathways. The references include:

- Journal: Cardiovasc Res 83:778-84 (2009)
DOI:10.1093/cvr/cvp163
PMID:11117527
Authors: Lazennec G, Canaple L, Saugy D, Wahli W
Title: Activation of peroxisome proliferator-activated receptors (PPARs) by their ligands and protein kinase A activators.
Journal: Mol endocrinol 14:1962-75 (2000)
DOI:10.1218/mend.14.12.0575
- Reference: PMID:16460276
Authors: Murthy KS
Title: Signaling for contraction and relaxation in smooth muscle of the gut.
Journal: Annu Rev Physiol 68:345-74 (2006)
DOI:10.1146/annurev.physiol.68.040504.094707
- Reference: PMID:18064658
Authors: Birukova AA, Zagranichnaya T, Alekseeva E, Bokoch GM, Birukov KG
Title: Epac/Rap and PKA are novel mechanisms of AMP-induced Rac-mediated pulmonary endothelial barrier protection.
Journal: J cell Physiol 215:715-24 (2008)
DOI:10.1002/jcp.21354
- Reference: PMID:9204709
Authors: Houslay MD, Milligan G
Title: Tailoring cAMP-signalling responses through isoform multiplicity.
Journal: Trends Biochem Sci 22:217-24 (1997)
DOI:10.1016/S0968-0004(97)01050-5

Related pathway: map00071 Fatty acid degradation
map04015 Rap1 signaling pathway
map04020 Calcium signaling pathway
map04151 PI3K-Akt signaling pathway
map04210 Apoptosis
map04260 Cardiac muscle contraction
map04340 Hedgehog signaling pathway
map04720 Long-term potentiation
map04740 Olfactory transduction
map04972 Pancreatic secretion
map04976 Bile secretion

KEGG pathway: K004024

DBGET integrated database retrieval system



The screenshot shows the AmiGO 2: Term Details for 'cAMP-mediated signaling' (GO:0019933). The page displays a list of annotations and related terms. The annotations include:

- Gene/product: pde6c
Phosphodiesterase
Annotation qualifier: cAMP-mediated signaling
GO class (direct): cAMP-mediated signaling
Annotation extension: GO_Central
Contributor: Oryzias latipes
Organism: IBA
Evidence: PANTHER:PTN001682918
Evidence with: CGD:CAL0000178266
PANTHER family: cyclic nucleotide phosphodiesterase pthr11347
Type: protein
- Gene/product: pde1b
Phosphodiesterase
Annotation qualifier: cAMP-mediated signaling
GO class (direct): cAMP-mediated signaling
Annotation extension: GO_Central
Contributor: Oryzias latipes
Organism: IBA
Evidence: PANTHER:PTN001682918
Evidence with: CGD:CAL0000178266
PANTHER family: cyclic nucleotide phosphodiesterase pthr11347
Type: protein
- Gene/product: 20197028
PDEase domain-containing protein
Annotation qualifier: cAMP-mediated signaling
GO class (direct): cAMP-mediated signaling
Annotation extension: GO_Central
Contributor: Helobdella robusta
Organism: IBA
Evidence: PANTHER:PTN001682918
Evidence with: CGD:CAL0000178266
PANTHER family: cyclic nucleotide phosphodiesterase pthr11347
Type: protein
- Gene/product: pde4d
Phosphodiesterase
Annotation qualifier: cAMP-mediated signaling
GO class (direct): cAMP-mediated signaling
Annotation extension: GO_Central
Contributor: Oryzias latipes
Organism: IBA
Evidence: PANTHER:PTN001682918
Evidence with: CGD:CAL0000178266
PANTHER family: cyclic nucleotide phosphodiesterase pthr11347
Type: protein
- Gene/product: 20196533
PDEase domain-containing protein
Annotation qualifier: cAMP-mediated signaling
GO class (direct): cAMP-mediated signaling
Annotation extension: GO_Central
Contributor: Helobdella robusta
Organism: IBA
Evidence: PANTHER:PTN001682918
Evidence with: CGD:CAL0000178266
PANTHER family: cyclic nucleotide phosphodiesterase pthr11347
Type: protein
- Gene/product: PDE10A
Phosphodiesterase
Annotation qualifier: cAMP-mediated signaling
GO class (direct): cAMP-mediated signaling
Annotation extension: GO_Central
Contributor: Felis catus
Organism: IBA
Evidence: PANTHER:PTN001682918
Evidence with: CGD:CAL0000178266
PANTHER family: cyclic nucleotide phosphodiesterase pthr11347
Type: protein
- Gene/product: PDE6A
Phosphodiesterase
Annotation qualifier: cAMP-mediated signaling
GO class (direct): cAMP-mediated signaling
Annotation extension: GO_Central
Contributor: Felis catus
Organism: IBA
Evidence: PANTHER:PTN001682918
Evidence with: CGD:CAL0000178266
PANTHER family: cyclic nucleotide phosphodiesterase pthr11347
Type: protein
- Gene/product: PDE2A
Phosphodiesterase
Annotation qualifier: cAMP-mediated signaling
GO class (direct): cAMP-mediated signaling
Annotation extension: GO_Central
Contributor: Felis catus
Organism: IBA
Evidence: PANTHER:PTN001682918
Evidence with: CGD:CAL0000178266
PANTHER family: cyclic nucleotide phosphodiesterase pthr11347
Type: protein

COMPARE HOW SEQUENCE DATA IS STRUCTURED IN NCBI vs UniProt

S.NO.	NCBI	UniProt
1.	It focuses on nucleotide sequences (DNA/RNA) and related genomic information.	It focuses on protein sequences and functional information.
2.	Some databases involved are: GenBank- Primary nucleotide sequence database. RefSeq- A curated, non-redundant database of reference sequences. NCBI Structure- It has experimentally resolved structures of proteins and nucleotides derived from PDB.	Some databases involved are: UniProtKB- A curated database of protein sequences and functional annotations. UniRef- Non-redundant sequence clusters at different levels of sequence identity. UniParc- A comprehensive archive of all known protein sequences.
3.	It uses variety of formats for storing sequence data, including FASTA and GenBank formats.	It uses an ASCII flat file format for its main distribution.
4.	It maintains the Sequence Read Archive for storing and analyzing Next-Generation Sequencing data.	Entries are structured to include sequence information, functional annotations, cross-references to other databases, and experimental data.
5.	It provides a comprehensive resource for nucleotide sequence data, genomic information, and related tools and services.	It provides a comprehensive and integrated resource for protein sequence and serves as a central hub for protein knowledge.
6.	It assigns unique identifiers to each sequence, which can be used to retrieve and manipulate the sequence data.	It assigns unique identifiers to each protein, which can be used to retrieve and manipulate the protein data.
7.	It provides information on sequence features, such as coding regions, non-coding regions, and regulatory elements.	It provides information on protein features, such as domains, motifs, and post-translational modifications.
8.	It offers tools for performing sequence alignments, which can be used to compare and analyze sequences.	It offers annotations on sequences, which can include information on protein function, structure and interactions.